

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 03

Date: 07.05.2026

Title: Interfacing Sensors with ESP32 in Wokwi

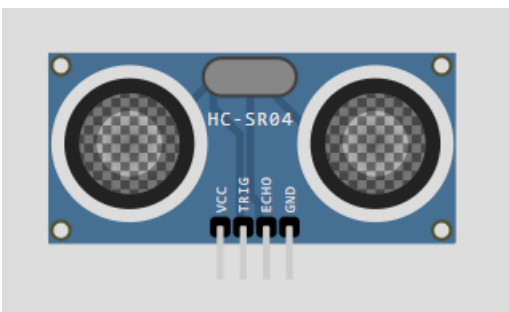
Aims: Learn how to correctly connect various sensors to the ESP32 Devkit in the Wokwi online simulator. This lab sheet focuses only on hardware connections (wiring) and includes a brief explanation of each sensor's purpose.

Tasks:

- Open wokwi.com and start a new project with ESP32 DevKit.
- Use the + button (Add Part) to search and add each sensor.
- Connect using wires (click and drag from pin to pin).
- Power: Most sensors use 3.3V or 5V (check individual notes). ESP32 GPIO pins are 3.3V logic.
- Always connect GND properly. Use recommended pins where possible for easy compatibility.

1. HC-SR04 (Ultrasonic Distance Sensor)

This sensor measures the distance to an object without physical contact by emitting ultrasonic sound waves and calculating the time it takes for the echo to return. It is commonly used in robotics for obstacle avoidance, parking assistance, level measurement, and interactive projects.



Pins on Sensor	Pins on ESP32
VCC	5V
TRIG	GPIO 5
ECHO	GPIO 18
GND	GND

2. DHT22 (Digital Humidity & Temperature Sensor)

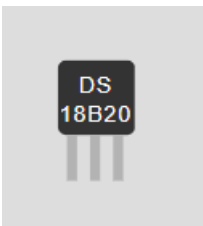
It measures ambient temperature and relative humidity with good accuracy. Widely used in weather stations, environmental monitoring, HVAC systems, and smart home automation.



Pins on Sensor	Pins on ESP32
VCC	3.3V
SDA	GPIO 4
NC	-
GND	GND

3. DS18B20 (One-Wire Digital Temperature Sensor)

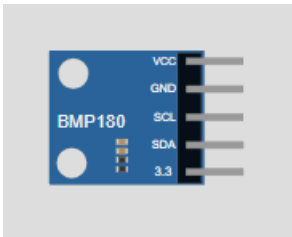
Provides precise digital temperature readings. Ideal for applications requiring accurate temperature monitoring (e.g., incubators, aquariums, industrial process control, or multiple-point temperature sensing) because multiple sensors can share one wire.



Pins on Sensor	Pins on ESP32
GND	GND
DQ	GPIO 4
VCC	3.3V

4. BMP180 (Barometric Pressure & Temperature Sensor)

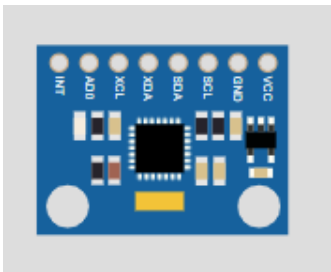
Measures atmospheric pressure and temperature. It can also calculate altitude. Commonly used in weather forecasting, altitude tracking (drones, hiking devices), and environmental monitoring.



Pins on Sensor	Pins on ESP32
VCC	3.3V
GND	GND
SCL	GPIO 22
SDA	GPIO 21
3.3V	-

5. MPU6050 (3-Axis Accelerometer + Gyroscope + Temperature)

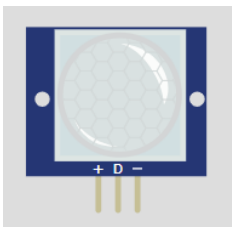
Detects motion, orientation, acceleration, and rotation. Essential for robotics, drones, self-balancing devices, gesture control, gaming controllers, and motion tracking applications.



Pins on Sensor	Pins on ESP32
INT	-
AD0	-
XCL	-
XDA	-
SDA	GPIO 21
SCL	GPIO 22
GND	GND
VCC	3.3V

6. PIR Motion Sensor (Passive Infrared)

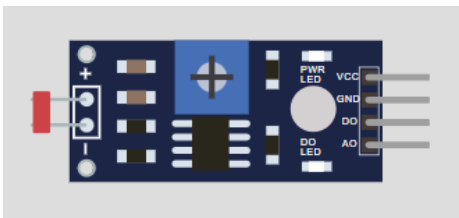
Detects movement of people or animals by sensing changes in infrared (heat) radiation. Primarily used in security alarms, automatic lighting, and presence detection systems. Digital output is HIGH on motion.



Pins on Sensor	Pins on ESP32
VCC	3.3V
OUT	GPIO 5
GND	GND

7. Photoresistor (LDR) Light Sensor

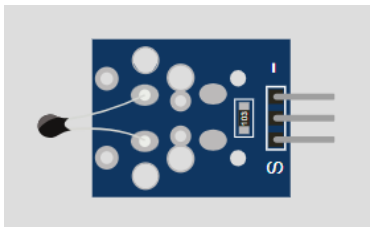
Detects light intensity by changing its resistance based on light levels. Used in automatic street lights, night lamps, camera exposure control, and light-activated systems.



Pins on Sensor	Pins on ESP32
VCC	3.3V
GND	GND
DO	-
AO	GPIO 34

8. Analog Temperature Sensor (NTC Thermistor)

Measures temperature by changing resistance with heat (Negative Temperature Coefficient). Used for temperature monitoring and control in appliances, battery management, and general environmental sensing.



Pins on Sensor	Pins on ESP32
GND	GND
VCC	3.3V
OUT	GPIO 34

9. MQ2 Gas Sensor

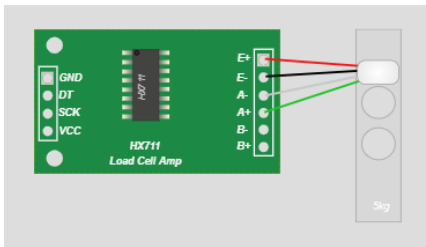
Detects combustible gases such as LPG, methane, propane, and smoke. Commonly used in gas leakage alarms, air quality monitors, and safety systems.



Pins on Sensor	Pins on ESP32
AOUT	GPIO 34
DOUT	-
GND	GND
VCC	5V

10. HX711 Load Cell Amplifier

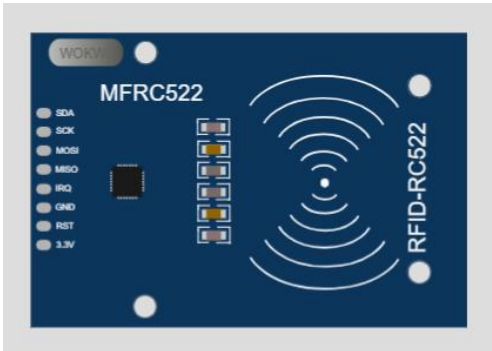
Amplifies tiny signals from load cells (strain gauges) to accurately measure weight. Used in digital scales, industrial weighing systems, and force/weight monitoring projects.



Pins on Sensor	Pins on ESP32
GND	GND
DT	GPIO 16
SCK	GPIO 17
VCC	5V

11. MFRC522 RFID/NFC Reader

Reads and writes RFID/NFC cards and tags for identification. Used in access control systems, attendance tracking, contactless payments, and inventory management.



Pins on Sensor	Pins on ESP32
SDA	GPIO 5
SCK	GPIO 18
MOSI	GPIO 23
MISO	GPIO 19
IRQ	-
GND	GND
RST	GPIO 21
3.3V	3.3V